Vélib' project: why we collected the data this way

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Our question

After the feedback we narrowed it to one thing: **how often is a Vélib' user left without a usable station within 300 m, and is it the same everywhere in the network?** We start from that question and only collect what we need to answer it.

What we need, and where it comes from

- **The state of every station over time.** The main data, collected by us from the public Vélib' GBFS feed every 5 minutes, plus the station list (names, coordinates, capacity) every hour.
- **Commune boundaries** (IGN ADMIN-EXPRESS), to know which area each station is in: the feed only gives coordinates. We use communes and not only Paris arrondissements, because a large part of the network is outside Paris.
- **Population by commune** (INSEE), so we can talk about how many people are affected and not only how many stations. Both are downloaded once, not collected over time.

We dropped the bike counters, the commuting analysis, the weather and the school holidays. The counters cannot prove that a station is broken, which is what the feedback pointed out, and the other three tell us *when* problems happen while our question is about *where*. We can still answer the question without them, so we do not keep them.

Why every 5 minutes

Two things are easy to mix up: how often a station updates its own state decides what we can see, while the delay before something shows up in the feed only shifts our timestamps and cannot be measured from the feed alone. In our data the gap between our capture and the station's own last_reported is 31 to 35 minutes on every day, and 91.6% of our rows repeat a report we had already captured, so we poll much faster than the stations report. We keep 5 minutes anyway: we then see a new report at most 5 minutes after it appears, without guessing the stations' rhythm, and nine days only cost 99 MB. The limit is that we cannot look at anything shorter than half an hour, and our timestamps say when the system knew, not when it happened.

What we collected, and how we store it

From 20.09 at 21:05 UTC to 29.09 at around 07:25 UTC we captured 2,428 status files and 203 station files, which gives 3,686,536 rows once flattened. Every complete day has all 288 status captures and all 24 station captures, so 100% coverage, with no failed request, no gap longer than 10 minutes and all 2,631 files opening correctly. We know this because the collector logs every attempt, even a failed one, so a missing capture shows up instead of disappearing silently. The first and last days look incomplete only because they are the start time and the snapshot moment. We keep these nine days in three forms and they weigh very differently: 99 MB as raw gzipped JSON, 71 MB as one JSON file per day, about 10 MB as Parquet partitioned by date. These are our first numbers for the storage comparison, and we will add ingestion and query times before choosing.

Two machines, and why they are not laptops

The feed only publishes the current state, so anything we miss is lost for good. We run the collection on two always-on cloud VMs (Azure, Denmark East) rather than our own laptops: a laptop sleeps, moves, loses its connection and gets closed during a lecture. Two machines with the same script give us three things: if one dies we still have the period, we can measure our own reliability by comparing what each caught, and we can check the feed by seeing whether both read the same values at the same minute. It is not two datasets, since they watch the same feed: we merge them by removing duplicated reports. Everything above comes from the first machine; the comparison is not finished yet.

How we define a failure, and why 300 m

A station is empty when num_bikes_available is 0, full when num_docks_available is 0, and out of service when is_renting, is_returning or is_installed is 0. We keep those separate instead of adding them into one "failure" number, because an empty station can be working perfectly. We only count a failure if it lasts at least two captures in a row. Another station is a real alternative if it is within 300 m and not failing at the same moment; a station is isolated if no other station is within 300 m.

The 300 m is not our invention: it is the spacing Vélib' Métropole advertises for its own network. We took it as a starting point and checked it against the station coordinates. Across the 1,519 stations, 39.0% have another station within 200 m, 68.0% within 300 m and 88.5% within 500 m, with a median distance to the nearest station of 231 m. We keep 300 m because it separates things usefully: at 200 m almost two thirds of stations would be called isolated and the word stops meaning anything, while at 500 m we would be pretending a six or seven minute walk is a real alternative. We measure station to station because our user is already standing at a station when the problem happens. These are straight-line distances, so the real walk is longer and our isolation counts are optimistic.

Limitations and sources of bias: what we cannot claim

- A station only exists for us when the feed talks about it, so we cannot tell a broken station from one that stopped reporting. Hence "out of service" and never "broken".
- Because reports reach us with about 33 minutes of delay, short problems are probably invisible: we undercount failures rather than just adding noise.
- Nine days at the end of September is not a whole year: no winter, no strike, no big event.
- Around a third of the stations are outside Paris and are much more spread out, but the exact count depends on where the boundary is drawn. We fix it when we join the commune boundaries.
- Our script saves whatever the server sends, even if the request failed, so in theory a bad file could look like a good one. It did not happen here (no failed request, every file opens, all the same size) but we checked afterwards instead of preventing it. We fix the script after this submission.
- Some stations report numbers that do not add up, two have no capacity and a few are frozen for days: this goes to the cleaning step.

What we know we still have to settle

- Our episode rule counts two captures in a row, but stations only report every ~33 minutes, so two captures often show the same unchanged report. We will redefine it on distinct last_reported values and check the effect.
- We report the average delay but not how much it varies. If the delay moves, our episode durations are noisy and not only shifted.
- Our log is written by the collector itself, so it cannot record that it never ran; the second machine tests that. The full quality assessment of the raw files is due at the next checkpoint.

Legal side and reproducibility

Public open data from Vélib' Métropole / Smovengo under the ODbL licence, credited in our README. It describes stations and not people, so there is no personal data to protect, and our 312 requests a day on a public file bypass nothing. Because the feed only shows the present, someone can reproduce our method but not our data: our scripts let anyone start their own archive, never rebuild ours. We keep the raw files and will deposit the archive publicly when collection ends, so it outlives the machines.